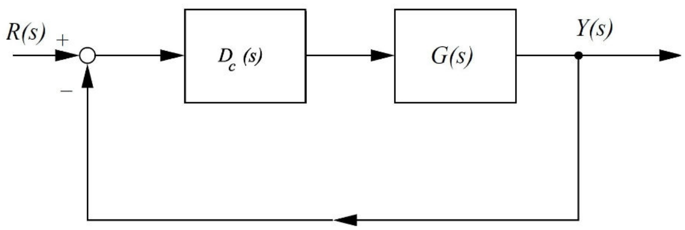
Homework 2

requiements:

- the phase margin must be at least 65∘ and
- the settling time within a $\pm 2\%$ band must be less than 2 seconds, in the case of a step input,

while

- the steady state error of the system in the case of a unit ramp input should not exceed 0.5.




```
clearvars, clc, close all
```

The plant is: $G(s) = \dfrac{4}{s(s+10)(s+2)}$

```
Gpoles = [0 -10 -2]
```

```
Gpoles = 1×3
     0   -10    -2
```

```
G = zpk([],Gpoles,4.0)
```

```
G =
 
         4
  --------------
  s (s+10) (s+2)
 
Continuous-time zero/pole/gain model.
Model Properties
```

To start off this design, the steady state error is calculated in terms of the ramp error constant K_v and thus a required value of is established:

$$e_{ss} = \frac{A}{K_v} \leq 0.5$$

$$K_v \geq 2$$

It is important to realize that this value represents the required K_v .

The actual K_v will depend on the gain chosen by the designer. If for instance the designer chooses to start his design by selecting $D_c(s) = 1$ and he/she then calculates the actual K_v. Here $K_v = \frac{4}{20} = 0.2$, is small by a factor of 10. So we can put a gain $K_1 = 10$ in series with the plant.

```
K1 = 10
```

```
K1 =
10
```

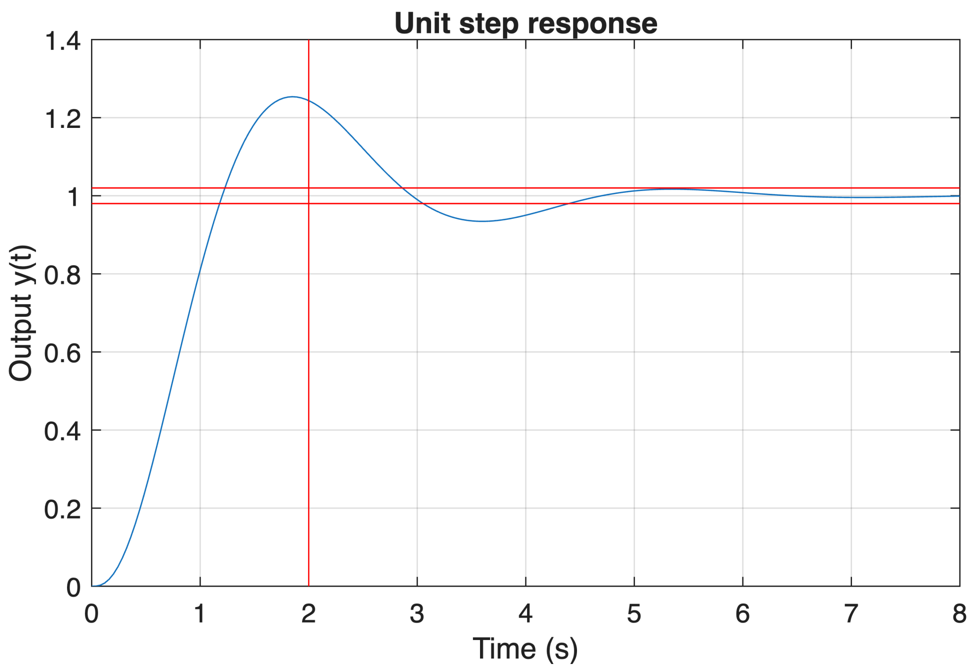
```
% kvcheck: NJ Theron's function to calculate
% velocity error constant
T = feedback(K1*G,1)
```

```
T =

                40
  --------------------------------
  (s+10.45) (s^2 + 1.547s + 3.827)

Continuous-time zero/pole/gain model.
Model Properties
```

```
tt = linspace(0.0,8.0,200);
y = step(T,tt);
figure(1)
plot(tt,y)
hold on
%plot([tt(1), tt(end)],[1.05, 1.05],'g')
plot([tt(1), tt(end)],[1.02, 1.02],'r')
plot([tt(1), tt(end)],[0.98, 0.98],'r')
plot([2,2],[0,1.4],'r')
grid on
xlabel('Time (s)')
ylabel('Output y(t)')
title('Unit step response')
```



Now we plot the Bode diagram of the uncompensated system.

```
figure(2)
omega = logspace(0,2,400);
[mag1,phase1] = bode(K1*G,omega);
mag1 = squeeze(mag1);
phase1 = squeeze(phase1);
subplot(2,1,1),semilogx(omega,20*log10(mag1))
xlabel('Angular frequency \omega (rad/s)'),ylabel('20log|G(\omega)|')
hold on
grid on
title('Plant (open loop) transfer function bode diagram')
subplot(2,1,2),semilogx(omega,phase1)
xlabel('Angular frequency \omega (rad/s)'),ylabel('\phi')
grid on
[Gm,Pm,omegcp,omegcg] = margin(K1*G);
dBGm = 20*log10(Gm);
disp(['System phase margin (degrees): ',num2str(Pm)])
```

```
System phase margin (degrees): 43.2099
```

```
disp(['System cross over angular frequency (rad/s): ',num2str(omegcg)])
```

```
System cross over angular frequency (rad/s): 1.5587
```

```
disp(['System gain margin (dB): ',num2str(dBGm)])
```

```
System gain margin (dB): 15.563
```

```
disp(['System Phase Crossover Frequency (rad/s):',num2str(omegcp)])
```

```
System Phase Crossover Frequency (rad/s):4.4721
```

We therefore need to turn to step 3: Allowing for a small amount of safety, determine the necessary additional phase lead $\phi_{\max}$.

Since the phase margin of the gain adjusted uncompensated system is about $43°$ degrees but we need $65°$, we would like to add about 22 degrees of additional phase to the phase diagram at the angular frequency where the compensated loop transfer function logarithmic gain curve will drop through 0 dB. It is

suggested that we add something as a measure of safety, so, I decided to start off my design iterations with $\phi_{\max} = 32°$.

From Franklin Eq.6.40

$$\alpha = \frac{1 - \sin\phi_{\max}}{1 + \sin\phi_{\max}}$$

```
phi = deg2rad(37)
```

```
phi =
0.6458
```

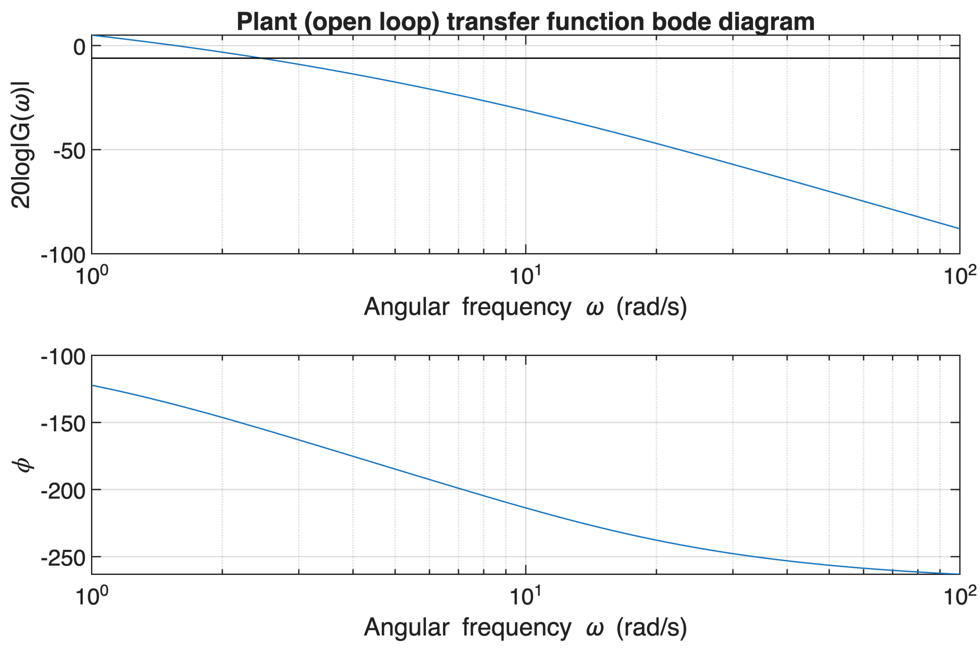
```
alpha = (1-sin(phi))/(1+sin(phi))
```

```
alpha =
0.2486
```

```
disp("To have phi_max = 32 degrees, we need alpha to be "+num2str(alpha))
```

```
To have phi_max = 32 degrees, we need alpha to be 0.24858
```

```
bb = 10*log10(alpha);
figure(2)
subplot(2,1,1),semilogx([omega(1), omega(end)],[bb, bb],'k')
```



```

```

From Bode diagram, the $\omega_{\max}$ is read off by 2.46 rad/s.

$$T_D = \frac{1}{\omega_{\max}\sqrt{\alpha}}$$

and Then the compensator is calculated as: $D_C = \dfrac{T_D s + 1}{\alpha\, T_D s + 1}$

```
%omegam = 2.46 % read off the Bode diagram in figure 2, for phi_max = 37
degrees
% below: read off the Bode diagram in figure 2, for phi_max = 37, but then
% slightly higher.
omegam = 3.6
```

```
omegam =
3.6000
```

```
%omegam = ginput(1);
%omegam = omegam(1);
disp("Compensator geometric mean frequency omegam = "+num2str(omegam)...
+" rad/s")
```

```
Compensator geometric mean frequency omegam = 3.6 rad/s
```

```
TD = 1/omegam/sqrt(alpha);
Dc = tf([TD, 1],[alpha*TD, 1])
```

```
Dc =

  0.5571 s + 1
  ------------
  0.1385 s + 1

Continuous-time transfer function.
Model Properties
```

```
disp("Lead compensator zero = -"+num2str(1/TD)+"rad/s")
```

```
Lead compensator zero = -1.7949rad/s
```

```
disp("Lead compensator pole = -"+num2str(1/(alpha*TD))+"rad/s")
```

```
Lead compensator pole = -7.2205rad/s
```

```
disp('Compensated loop transfer function:')
```

```
Compensated loop transfer function:
```

```
Gt = Dc*K1*G
```

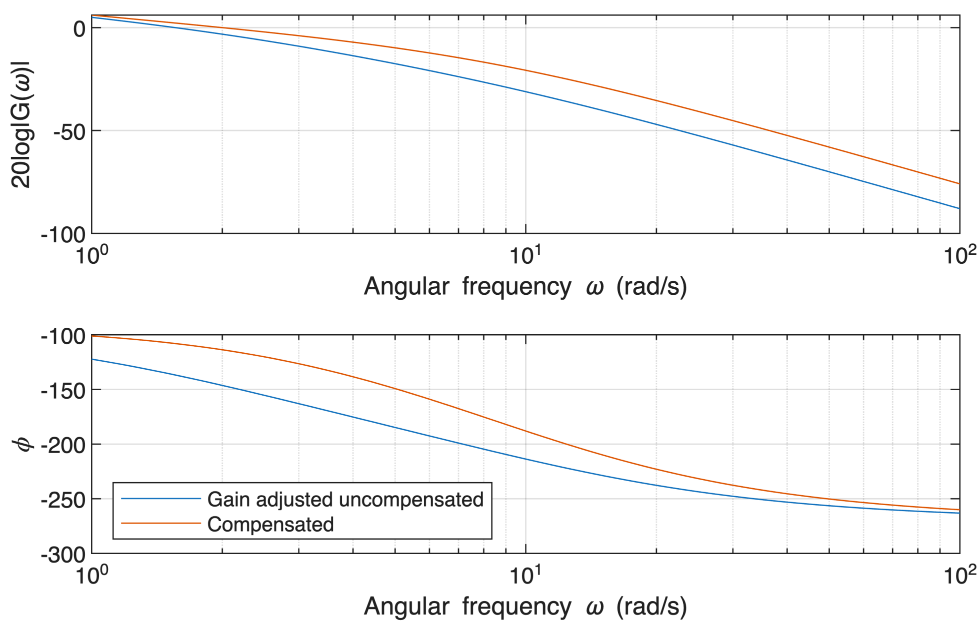
```
Gt =

     160.91 (s+1.795)
  -----------------------
```

```
  s (s+7.22) (s+10) (s+2)

Continuous-time zero/pole/gain model.
Model Properties
```

```
figure(3)
title('Loop transfer functions Bode diagrams')
[mag2,phase2] = bode(Gt,omega);
mag2 = squeeze(mag2);
phase2 = squeeze(phase2);
subplot(2,1,1)
semilogx(omega,20*log10(mag1))
hold on
semilogx(omega,20*log10(mag2))
hold off
grid on
xlabel('Angular frequency \omega (rad/s)'),ylabel('20log|G(\omega)|')
subplot(2,1,2)
semilogx(omega,phase1)
hold on
semilogx(omega,phase2)
hold off
grid on
xlabel('Angular frequency \omega (rad/s)'),ylabel('\phi')
legend('Gain adjusted uncompensated','Compensated','Location','southwest')
```




```
[Gm,Pm,omegcp,omegcg] = margin(Gt);
dBGm = 20*log10(Gm);
```

```
disp(['Compensated system phase margin (degrees): ',num2str(Pm)])
```

```
Compensated system phase margin (degrees): 66.2905
```

```
disp(['Compensated system cross over angular frequency (rad/s): ',...
num2str(omegcg)])
```

```
Compensated system cross over angular frequency (rad/s): 2.0009
```

```
disp(['Compensated system gain margin (dB): ',num2str(dBGm)])
```

```
Compensated system gain margin (dB): 18.2018
```

```
disp(['Compensated system gain margin cross over angular frequency (rad/s):
'...
,num2str(omegcp)])
```

```
Compensated system gain margin cross over angular frequency (rad/s): 8.6935
```

Then For the first $\omega_{\max} = 2.7$, the PM didn't meet the requiement, we do another iteration, increase the $\omega_{\max} = 3$, The PM meet the requiements

```
disp("Compensated system transfer function"+...
"(i.e., the closed loop transfer function):")
```

```
Compensated system transfer function(i.e., the closed loop transfer function):
```

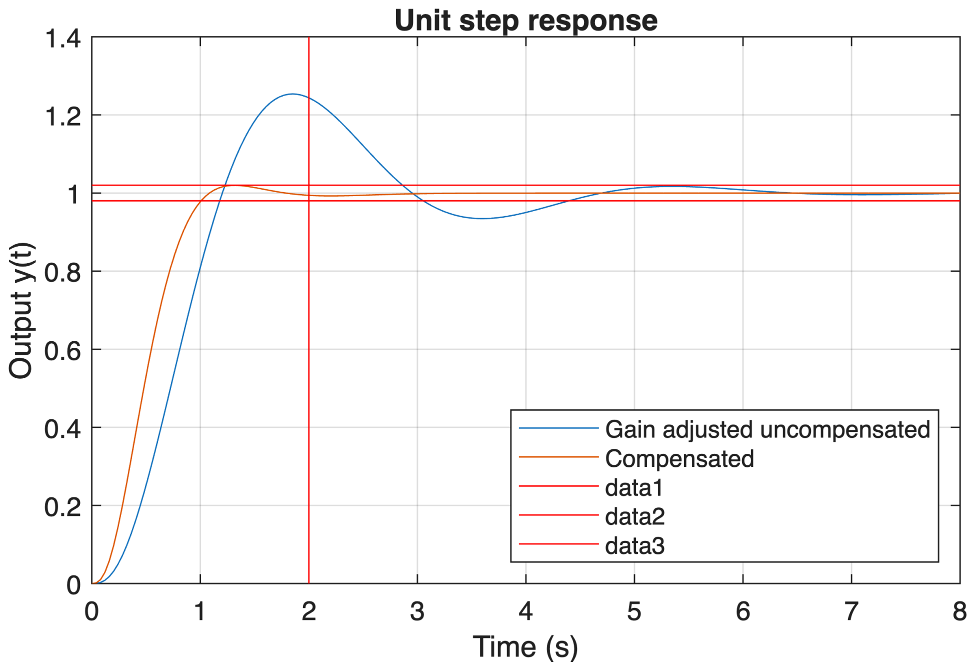
```
T2 = feedback(Gt,1)
```

```
T2 =

             160.91 (s+1.795)
  ------------------------------------------
  (s+12.49) (s+1.612) (s^2 + 5.117s + 14.34)

Continuous-time zero/pole/gain model.
Model Properties
```

```
y2 = step(T2,tt);
figure(4)
plot(tt,y,tt,y2)
legend('Gain adjusted uncompensated','Compensated','location','southeast')
hold on
plot([tt(1), tt(end)],[1.02, 1.02],'r')
plot([tt(1), tt(end)],[0.98, 0.98],'r')
plot([2, 2],[0.0, 1.4],'r')
grid on
xlabel('Time (s)')
ylabel('Output y(t)')
title('Unit step response')
hold off
```


Unit step response
1.4
1.2
1
0.8
0.6
0.4
0.2
0
Output y(t)
0
1
2
3
4
5
6
7
8
Time (s)
Gain adjusted uncompensated
Compensated
data1
data2
data3